

Name

- Show appropriate work.
 - Appropriate work for a linear ODE includes the integrating factor and some intermediate steps. Write down any linear systems you solve. Do not solve if asked for an Augmented Matrix.
 - Appropriate work for a separable ODE includes the separated form, some intermediate steps for integration, a general solution, and if appropriate a clear substitution for ICs. Do not solve for y .
 - Appropriate work for an exact ODE includes an explicit test for exactness, and details of each step of the process including a general solution and if appropriate a clear substitution for ICs. Do not solve for y .
 - Appropriate work for a linear system $y' = Ay + b$ includes a fully defined real complementary solution, appropriate undetermined coefficient forms, and some work indicating how needed linear systems are being constructed. Write down any linear systems you solve. Do not solve if asked for an Augmented Matrix.
 - Appropriate work for a higher order constant coefficient linear ODE includes the Characteristic Polynomial, appropriate undetermined coefficient forms, and some work indicating how linear systems are being constructed. Write down any linear systems you solve. Do not solve if asked for an Augmented Matrix.
- Explicitly real solutions are required.
- Computations
 - You are given Mathematica output for the the eigenvalues and eigenvectors A when asked to solve $y' = Ay + b$. Do not attempt to recompute eigenvalues etc.
 - You are given a factored form of the characteristic polynomial for polynomials of degree higher than 2. You need to compute roots of quadratics.
 - You know several ways to build an Augmented Matrix (AM) out of vectors and smaller matrices.
 - The matrix for the linear system $\sum_{i=1}^n c_i v_i = b$ is $A \cdot c = b$ where the columns of $A = (v_1 \dots v_n)$ are the vectors $v_1, v_2, \dots$ and an AM is $(A \mid b)$.
 - An AM for the linear system
$$\begin{aligned} A_{11}c_1 + A_{12}c_2 &= b_1 \\ A_{21}c_1 + A_{22}c_2 &= b_2 \end{aligned}$$
 is $\left(\begin{array}{cc|c} A_{11} & A_{12} & b_1 \\ A_{21} & A_{22} & b_2 \end{array} \right)$.
 - I_n is the $n \times n$ identity matrix with 1s on the diagonal and zeroes everywhere else.
 - Labelling and using eigenvalues/eigenvectors, roots etc. appropriately is a good idea.
- The exam has 7 questions on six pages. 6 of the questions are worth 7 points and one of the questions is worth 8 points.
 - There are 2 or 3 questions on the material from exam 1
 - There are 2 or 3 questions on the material from exam 2
 - There are 2 questions on the material from exam 3

④

$$\cos(t) y' - \sin(t) y = 3 \cos(t)$$

$$y(\pi/2) = 3$$

$$y' - \frac{\sin(t)}{\cos(t)} y = 3$$

$$v(t) = e^{\int -\sin(t)/\cos(t) dt}$$

$$u = \cos(t)$$

$$du = -\sin(t) dt$$

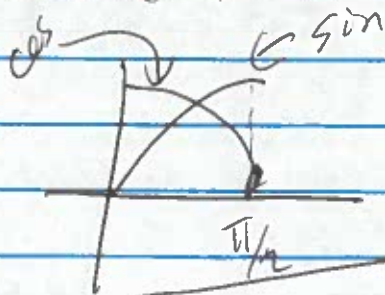
$$v(t) = e^{\ln(\cos(t))} = \cos(t)$$

$$(\cos(t) y)' = 3 \cos(t)$$

$$\cos(t) y = 3 \int \cos(t) dt + C$$

$$\cos(t) y = 3 \sin(t) + C$$

$$\cos(\pi/2) 3 = 3 \sin(\pi/2) + C$$



$$0 = 3 + C \quad | \quad C = -3$$

$$y = 3 \frac{\sin(t)}{\cos(t)} + \frac{-3}{\cos(t)}$$

⑥

$$dy/dt = 4t e^{t^2} \cdot e^y$$

$$y(1) = 2$$

$$\int e^{-y} dy = \int 4t e^{t^2} dt + C$$

$$\boxed{\begin{array}{l} u = t^2 \\ du = 2t dt \end{array}}$$

$$-e^{-y} = \int 2 e^u du + C$$

Gen
sol

$$\boxed{-e^{-y} = 2e^{t^2} + C}$$

$$-e^{-2} = 2e^{1^2} + C$$

$$\boxed{C = -\frac{1}{2} - 2e}$$

$$\boxed{-e^{-y} = 2e^{t^2} - \frac{1}{2} - 2e}$$

②

$$\left(\frac{1}{t} + \cos(y^2)\right) dt + (3y^2 - 2ty \sin(y^2)) dy = 0$$

M N

$y(1) = \sqrt{\pi}$

$$\frac{\partial M}{\partial y} = 0 - \sin(y^2) 2y \quad \frac{\partial N}{\partial t} = 0 - 2y \sin(y^2)$$

Exact 😊

$$\frac{\partial s}{\partial t} = \frac{1}{t} + \cos(y^2) \Rightarrow s = \ln(t) + t \cos(y^2) + h(y)$$

$$\begin{aligned} \frac{\partial s}{\partial y} &= 0 + t(-\sin(y^2) 2y) + h'(y) \\ &= -2ty \sin(y^2) + 3y^2 \end{aligned}$$

$$h'(y) = 3y^2 \quad h(y) = y^3$$

Gen sol $\ln(t) + t \cos(y^2) + y^3 = C$

$$\ln(1) + 1 \cos(\pi) + (\sqrt{\pi})^3 = C$$

$$0 - 1 + \pi^{3/2} = C$$

$$\boxed{\ln(t) + t \cos(y^2) + y^3 = -1 + \pi^{3/2}}$$

9/

$$(1 - y \sin(ty)) dt + (6y - t \sin(ty)) dy = 0$$

M N $y(1) = 1$

$$\frac{\partial M}{\partial y} = 0 - \sin(ty)$$

$$\frac{\partial N}{\partial t} = 0 - (1) \sin(ty)$$

$$\frac{\partial N}{\partial t} = 0 - (1) \sin(ty)$$

$$- t \cos(ty) \cdot y$$

same Exact 😊

$$\frac{\partial S}{\partial y} = 6y - t \sin(ty) \Rightarrow S = \int 6y dy - \int t \sin(ty) dy$$

$$S = 6 \frac{y^2}{2} - \int \sin(ty) t dy = 3y^2 - \int \sin(u) du + h(t)$$

$$\boxed{\begin{array}{l} u = ty \\ du = t dy \end{array}}$$

$$S = 3y^2 + \cos(ty) + h(t)$$

"m"

$$\frac{\partial S}{\partial t} = 0 + -\sin(ty)y + h'(t) = 1 - y \sin(ty)$$

$$h'(t) = 1 \quad h(t) = t$$

Gen sol

$$\boxed{3y^2 + \cos(ty) + t = C} \quad \boxed{y(1) = 1}$$

$$3 + \cos(1) + 1 = C$$

$$\boxed{C = 4 + \cos(1)}$$

Exam 2: Constant Coefficient Linear Systems of ODE

$$y' = Ay + b \text{ with } y(0) = y_0$$

1. Two parts

1.1. Write down an AM you would row reduce for the ODE $y' = Ay + b$ with $y(0) = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$

$$y_p = \begin{pmatrix} 3 + \cos(t) \\ 5 + \sin(t) \\ 6 + t \end{pmatrix} \text{ and } y_p(0) = \begin{pmatrix} 3+1 \\ 5+0 \\ 6+0 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

$$y_c = c_1 e^{2t} \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} + c_2 e^t \left(\cos(2t) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \sin(2t) \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right) + c_3 e^{2t} \left(\sin(2t) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \cos(2t) \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right)$$

$$\text{AM} = \left[\begin{array}{ccc|cc} -3 & 1 & 1 & 2 & -4 \\ 1 & 1 & 1 & 3 & -5 \\ 2 & 1 & -2 & 4 & -6 \end{array} \right]$$

$$y = y_c + y_p$$

$$\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = c_1 \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

$$y = y_c + y_p$$

1.2. Compute the solution of $y' = Ay + e^{-t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ where A is below

$$\text{In[*]:= MatrixForm[Eigensystem[A = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}]]]$$

Out[*]//MatrixForm=

$$\begin{pmatrix} 7 & 3 \\ \{1, 1\} & \{-1, 1\} \end{pmatrix}$$

2. Two parts

2.1. Write down an AM you would row reduce for the ODE $y' = Ay + b$ with $y(0) = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$$y_p = \begin{pmatrix} 2e^t + \sin(2t) \\ 5t + \cos(2t) \end{pmatrix} \text{ and } y_c = c_1 e^{3t} \begin{pmatrix} 1 \\ -3 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\text{AM} = \left[\begin{array}{cc|cc} & & & \\ & & & \end{array} \right]$$

2.2. Compute the solution of $y' = Ay + e^{6t} \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ where A is below

$$y_c = c_1 e^{7t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{3t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$y_p = e^{-t} \vec{a} \quad y_p' = -e^{-t} \vec{a}$$

$$-e^{-t} \vec{a} = A e^{-t} \vec{a} + e^{-t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$[-I - A] \vec{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

continued on sheet

Ex2

1.2

$$[-I - A] \vec{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\left[\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix} \right] \vec{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\left[\begin{array}{cc|c} -6 & -2 & 1 \\ -2 & -6 & 0 \end{array} \right] \Rightarrow -2a_1 - 6a_2 = 0$$

$$\boxed{a_1 = -3a_2}$$

$$-6a_1 - 2a_2 = 1$$

$$\Rightarrow -6(3a_2) - 2a_2 = 1 \Rightarrow -16a_2 = 1$$

$$\boxed{a_2 = -1/16}$$

$$\vec{y}_p = e^{-t} \begin{pmatrix} +3/16 \\ -1/16 \end{pmatrix}$$

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In[*]:= MatrixForm[Eigensystem[A =  $\begin{pmatrix} 6 & -2 & 0 \\ 2 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix}$ ]]
```

```
Out[*]//MatrixForm=
```

$$\left(\begin{array}{ccc} 6+2i & 6-2i & 5 \\ \{1+2i, 2-i, 1\} & \{1-2i, 2+i, 1\} & \{0, 0, 1\} \end{array} \right)$$

3. Two parts

3.1. Write down an AM you would row reduce for the ODE $y' = Ay + b$ with $y(0) = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

$$y_p = \begin{pmatrix} 7e^{5t} + 2\sin(3t) \\ 4 + \sin(3t) \end{pmatrix} \text{ and } y_c = c_1 e^{3t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^t \begin{pmatrix} 7 \\ 2 \end{pmatrix}$$

$$\text{AM} = \left(\begin{array}{c} \end{array} \right)$$

3.2. Compute the solution of $y' = Ay + \begin{pmatrix} 8 \\ 13 \\ 4 \end{pmatrix}$ where A is below

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In[*]:= MatrixForm[Eigensystem[A =  $\begin{pmatrix} 6 & 1 & 0 \\ 1 & 6 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ ]]
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Out[*]//MatrixForm=
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$$\left(\begin{array}{ccc} 7 & 5 & 1 \\ \{6, 6, 1\} & \{4, -4, 1\} & \{0, 0, 1\} \end{array} \right)$$

4. Two parts

4.1. Compute the solution of $y' = Ay + e^{5t} \begin{pmatrix} 8 \\ 0 \\ 8 \end{pmatrix}$ where A is below

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In[*]:= MatrixForm[Eigensystem[A =  $\begin{pmatrix} 6 & -1 & 0 \\ 1 & 6 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ ]]
```

```
Out[*]//MatrixForm=
```

$$\left(\begin{array}{ccc} 6+i & 6-i & 1 \\ \{5+i, 1-5i, 1\} & \{5-i, 1+5i, 1\} & \{0, 0, 1\} \end{array} \right)$$

4.2. Write down an AM you would row reduce for the solution from 4.1 with IC $y(0) = \begin{pmatrix} 3 \\ 1 \\ 8 \end{pmatrix}$

$$AM = \left(\begin{array}{c} \\ \\ \end{array} \right)$$

5. Two parts

5.1. Compute the solution of $y' = Ay$ where A is below

```
In[ ] = MatrixForm[Eigensystem[A =  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ ]]
```

```
Out[ ] // MatrixForm =
```

$$\left(\begin{array}{cc} 5 & 0 \\ \{1, 2\} & \{-2, 1\} \end{array} \right)$$

5.2. Write down an appropriate y_p form for $y' = Ay + \sin(2t)\begin{pmatrix} 1 \\ 4 \end{pmatrix}$. Write down a AM to find the coefficients

in y_p .

$$y(t) =$$

$$AM = \left(\begin{array}{c} \\ \\ \end{array} \right)$$

Exam 3: Higher Order Constant Coefficient Linear ODEs (with appropriate IC)

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_2 y'' + a_1 y' + a_0 y = b$$

There will be three 10 point questions on this material. Examples are below.

1. Two parts

1.1. Solve $y'' + 2y' - 3y = 10\sin(t)$

$$y =$$

1.2. Write down the AM you would solve to determine the constants of integration for $y(0) = 1$ and $y'(0) = 3$.

AM = $\left(\begin{array}{c} \\ \\ \end{array} \right)$

2. Two parts

2.1. Solve $y'' - 2y' + 5y = 10e^{2t}$

$y =$

2.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$

AM = $\left(\begin{array}{c} \\ \\ \end{array} \right)$

3. Two parts. Given

In[] := Expand[(r + 2)² (r - 1)]

Out[] :=

$$-4 + 3r^2 + r^3$$

3.1. Solve $y''' + 3y'' - 4y = 32e^{2t}$

$y =$

3.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$ and $y''(0) = 3$

AM = $\left(\begin{array}{c} \\ \\ \end{array} \right)$

4. Two parts. Given

In[]:= Expand[(r² + 4) (r - 1)]

Out[]:=
-4 + 4 r - r² + r³

4.1. Solve $y''' - y'' + 4y' - 4y = 32e^{-t}$

$$y =$$

4.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$ and $y''(0) = 3$

$$\text{AM} = \left(\right)$$

5. Two parts. Given

In[]:= Expand[(r + 1)³]

Out[]:=
1 + 3 r + 3 r² + r³

5.1. Solve $y''' + 3y'' + 3y' + y = 8e^t$

$$y =$$

5.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$ and $y''(0) = 3$

$$\text{AM} = \left(\right)$$

6. Two parts. Given

In[] := Expand[(r - 2)(r + 1)(r + 3)]

Out[] :=
 $-6 - 5r + 2r^2 + r^3$

6.1. Solve $y''' + 2y'' - 5y' - 6y = 3t$

$y =$

6.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -2$ and $y'(0) = -5$ and $y''(0) = 1$

AM = $\left(\begin{array}{c} \\ \\ \end{array} \right)$

7. Two parts. Given

In[] := Expand[(r - 2)²]

Out[] :=
 $4 - 4r + r^2$

7.1. Solve $y'' - 4y' + 4y = 0$

$y_c =$

7.2. Write down the form you would choose for a particular solution for $y'' - 4y' + 4y = 3\sin(2t)$ and the AM for the coefficients in your form

$y_p =$

AM = $\left(\begin{array}{c} \\ \\ \end{array} \right)$

Exam 1: Linear, Exact and Separable ODEs

1. Solve the ODE $3y'(t) + 4y(t) = -13e^{2t}$ with $y(0) = 2$

2. Solve the ODE $2t^2 y'(t) - 3ty(t) = -4t^7$ with $y(1) = -7$.

✓ 4. Solve the ODE $\cos(t)y' - \sin(t)y = 3\cos(t)$ with $y(\pi/2) = 3$. *Linear on sheet*

5. Solve the ODE $y' + 2t \tan(t^2)y = 1\cos(t^2)$ with $y(0) = 5$.

All with a box saying

$y(t) =$

✓ 6. Solve the ODE $y' = 4te^{t^2+y}$ with $y(1) = 2$.

on sheet

7. Solve the ODE $yy' = e^{y^2}t^2$ with $y(1) = 2$.

✓ 8. Solve the ODE $(1/t + \cos(y^2))dt + (3y^2 - 2ty \sin(y^2))dy = 0$ with $y(1) = \sqrt{\pi}$.

✓ 9. Solve the ODE $(1 - y \sin(ty))dt + (6y - t \sin(ty))dy = 0$ with $y(1) = 1$.

on sheets